Ref. and system	Stage	C1 clinical signal	C2 real X-ray validated	C3 task variables	C4 failure + handover	Total (of 8)	Reported result (n)
Target: all four criteria at 2		2	2	2	2	8	met by 0 of 21 autonomous rows
Real X-ray in the loop (12 autonomous rows)							
■ [6]* AI-driven robotic PCI	in vivo	2	2	2	1	7	correct commands 83.6–95.6% of 420 decisions; 3 pigs
■ [35]* VascularPilot3D	bench	2	1	2	1	6	25/25 (5 tasks × 5 trials)
■ [7] Diffusion policy, renal access	in vivo	2	2	1	1	6	in vivo 2/2 trials, 2 pigs
● [4]† Recurrent SAC, BCI access	bench	2	1	1	1	5	14/20
■ [13] Karstensen RL, porcine liver	ex vivo	2	2	1	0	5	9/30
■ [113]* VISTA microrobot (cross-domain)	in vivo	2	1	2	0	5	3 trials per condition (bench); rat run qualitative
● [1]*† TD-MPC2 world model	bench	2	1	1	0	4	17/25
● [2]*† PEF multi-task world model	bench	2	1	1	0	4	unseen phantom 18/30; 20/30 fine-tuned
■ [3]*† stEVE BasicWireNav, X-ray	bench	2	1	1	0	4	10/10 (first 10 targets)
■ [34] Modified COAST closed loop	bench	2	1	1	0	4	tip error 1.49, 2.13 mm, n = 2; one correction; airway phantom
■ [72]* COAST motion under X-ray	bench	2*	1*	1*	0*	4*	path error 2.87 mm; n not in abstract
■ [41] SplineFormer	bench	2	1	0	0	3	BCA 10/20; LCCA not achieved (n not stated)
Simulator video in the loop (not X-ray)							
■ [36] Dual-device perception-control	bench	0	0	2	1	3	63/65 runs without failure-rule events
Tip camera in the loop (no X-ray; listed for contrast)							
■ [39] Haptic vision, intracardiac	in vivo	0	0	2	2	4	89/90; 43/65; 79/83 (three tasks)
Overhead camera or EM tracking in the loop (not X-ray)							
■ [61] Online expert correction	bench	0	0	1	1	2	177/300 training episodes
■ [3]* stEVE BasicWireNav, camera	bench	0	0	1	0	1	97/100
● [54] HM-MARL	bench	0	0	1	0	1	right 7–10/10; left 0/10
● [42] Soft robotic guidewire IL	bench	0	0	1	0	1	15/18 novel; 16/18 rearranged
■ [48] Knowledge-driven visual guidance	bench	0	0	1	0	1	100% on LSA, LCA and BCA, 20 per target
■ [63] Siamese tip tracking on a robot	bench	0	0	1	0	1	tracking only; no success rate
■ [138] Chi et al. EM cannulation	bench	0	0	1	0	1	98.1% / 94.4% (n not extracted)
Shared control (LoA 1): not autonomous, not ranked							
■ [131] Bending-aware co-pilot (LoA 1)	in vivo	2	2	2	1	(7)	shared control; bending risk 239.8 → 80.5

Score: 0 not met or not described 1 partly met 2 met * card unverified or with pending fields; 2* = abstract-level score

Loop signal: ■ real fluoroscopy ■ camera or simulated fluoroscopy ■ tip camera ■ EM tracking ○ neurovascular □ other anatomy

† loop closed and scored through the [5] tip tracker (accuracy never measured)